Ethan Soh

APPLIED AI · AGENT ENGINEERING · AI SECURITY

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Looking for a 44-week internship from 12 April 2027 to 11 February 2028.

ABOUT ME

I'm a second-year Cybersecurity and Digital Forensics student at Singapore Polytechnic. Most of my time outside class goes into building things with AI agents and then trying to make them behave: I write the rules they follow, the checks that stop them, and the tests that prove the checks work. I hold Anthropic's Claude Certified Architect – Foundations (CCA-F) and ISC2 Certified in Cybersecurity, and I'm approved into Anthropic's Cyber Verification Program (dual-use security work), Jul 2026. I'm looking for an internship in applied AI, agent engineering or AI security.

WHAT I'VE BUILT

Agentroid · Public preview, Apache 2.0: github.com/Sohwhat6420/agentroid-mcp — Oct 2026 – present

– A TypeScript MCP server and Rust daemon that let Claude Code operate Android phones and emulators over adb. What I care about most is that the safety checks live in the daemon, not in a prompt: every adb command is blocked unless allowed, each device has to be on an allowlist, a screen guard blocks payments, factory resets and credential entry, every action carries a risk label from R0 to R5, and every decision goes to a SHA-256 hash-chained audit log. I red-teamed it: it refused or held 44 of 44 hostile test actions and allowed 12 of 12 ordinary ones, and the 4 known misses are published. It is a preview release, tried only with Claude Code; the daemon is Windows-only and drives emulators by default. There is also a Python SDK. It was written with AI coding agents under my design and review.

Human LLM Council · Open source, v0.1.0 preview; v1.0 in a draft pull request: github.com/Sohwhat6420/human-llm-council — 2026 – present

– A local "model of me" that reviews AI-agent output and blocks what I would reject. I built a blind benchmark from 1,887 of my real agent decisions (time-split; the agent's recommendation hidden and the options shuffled): copying the agent scores 71.5% overall but 0% on my 53 overrides, so a trained model has to beat 70.6% overall and 24.5% on overrides. 4-bit QLoRA on Qwen3-1.7B fits a 6 GB laptop GPU (3.34 GB peak, synthetic data only). The privacy pipeline pseudonymises, redacts and needs a signed review before training; a canary audit leaked 0 of 5 on synthetic data and its positive control caught 5 of 5. Transcripts are treated as untrusted input (OWASP LLM01). Designing next: weight-space programming (four LoRA adapters on one base model, combined by merging) and a nightly reflect, sleep and dream loop that adopts an update only if it beats the benchmark, and otherwise rolls back. No model has been trained on real data yet.

Multi-agent system I work in, and headless-kit · Tools open source (MIT): github.com/Sohwhat6420/headless-kit — Jul 2026 – present

– The setup I run my other work on. One Claude Code orchestrator splits work across parallel Claude and Gemini sessions on two laptops (some also drive phones); a router sends each job to the cheapest model pool that can do it, Claude reviews Gemini's output, and hooks enforce the written rules. A private guard that handles personal data refuses when it can't tell whether something is private. headless-kit (v0.2.0, CI on Windows and Linux) is the open-source part: zero-dependency tools and hooks for agents that run with no browser, sign-in or window. I direct the agents and review what they write; I don't pretend I typed every line.

Sidemic · Open source, v0.1.0 pre-release: github.com/Sohwhat6420/sidemic — 2026 – present

– Turns an Android phone into a wireless PC microphone with nothing installed on the phone (Rust, Tauri, Java). DeepFilterNet noise removal runs on the phone at 3.9 ms per 10 ms block, and the tray app idles at 8 MB, down from about 800 MB.

Awards and attendance system, The Boys' Brigade 5th Singapore Company — Apr 2026 – present

– Started as a Google Sheets tracker for the awards department (live 27 Apr 2026); from July, a Next.js, React and Supabase portal. It holds awards and attendance records and imports the training programme. I also built an AI assistant that only ever saw pseudonymised names and needed a person to confirm every write; I took it out in October after reviewing it, because the portal did not need that risk. We applied to the National Youth Council's Young ChangeMakers grant in September.

NUS-ISS Show Me Your Agents Hackathon 2026 · Team of four, with AWS and the Singapore Business Federation — Sep – Oct 2026

– We built an agentic AI solution to a real SME problem. We were not among the teams selected; I have the Letter of Participation.

SMALLER BUILDS, PUBLIC ON GITHUB

– **designyourtv and designyourwindows** a painted depth-clock lock screen, glass widgets and a remote-driven dock for Google TV, and the same iPhone-style depth clock on Windows as one 62 KB exe.

– **Profile banner** my GitHub profile banner is filmed from my TV's own home screen.

Hobby: android-film-simulations, Fujifilm looks on an Android phone's camera over adb, measured in ΔE2000 from about 490 shots.

Also in progress: Claude for HMOS Lite, to answer my coding agent from a Huawei watch (unofficial; not affiliated with Anthropic or Huawei).

WORK

Academic Content Assistant · Alchemy Education Studio · contract, remote — Feb 2025 – Oct 2026

– I write O-Level Biology material for a tuition centre: worksheets, revision papers, mock exams, answer keys and marking schemes, each checked for accuracy, syllabus fit and marking-point detail. I finished every paper for the 6093 Biology syllabus (Sec 3 and 4), and the centre uses them in lessons. I also built a pipeline that drafts each paper from a reference library and checks it.

Retail Assistant · Bee Cheng Hiang · part-time — Nov 2024 – Feb 2025

– Chinese New Year season: till, packing, restocking and customer questions, including Mandarin.

CERTIFICATIONS

Claude Certified Architect – Foundations (CCA-F)
Anthropic · issued Sep 2026, valid to Sep 2027
View on Credly

Certified in Cybersecurity (CC)
ISC2 · issued Jan 2026, valid to Jan 2029
View on Credly

Anthropic Academy, five courses (Claude 101, Claude Code 101, Claude Platform 101, Cowork, AI Fluency), Jul 2026. CPR + AED Provider, Singapore Red Cross, Oct 2026. Anthropic Cyber Verification Program, approved Jul 2026.

SCHOOLWORK I'M PROUD OF

- **Fortune's Descent** My Back-End Web Development CA2: a browser game on an Express API with JWT sign-in, validation and tests. I went through the code myself.
- **Offensive Security CA2** Group project: we assessed CVE-2025-47812 (Wing FTP Server) and scored it with CVSS, using Claude Pro.
- **Securing Microsoft Windows CA1** I did almost all of the hands-on Windows Server 2022 hardening, with Claude guiding: group policy, Active Directory, DNS, CIS audits.

COMPETITIONS AND EVENTS

- **Hack The Box Cyber Apocalypse CTF 2026** 30th of 6,744 teams, with Team Mythos Larpers. We solved all 136 challenges for 69,425 points.
- **Singapore Cyber Conquest 2026, Student Category** top 20 teams from the qualifiers; I went to the finals at Singapore International Cyber Week, 14 Oct.
- **DSTA BrainHack 2026, Cyber Defenders Discovery Camp** finalist in the Explorer category with team TUFF.
- **CyberBlitz 2026** I wrote binary-exploitation challenges for SP Gryphons, each with a Docker setup, a solver and a writeup.
- **National AI Student Challenge 2026, Workato track** participant, in a team of four. The brief was an MCP prototype.
- **TISC 2026 (CSIT)** Participant.

Also went to: Black Hat Asia 2026 (student pass), OB1 Conference 2026 (selected for SP's student sponsorship with STAR Labs SG), DEF CON Singapore 2026 (represented SP Gryphons), the OffSec Singapore Chapter meetup (May 2026).

LEADERSHIP AND VOLUNTEERING

Senior Cadet Lieutenant (Primer) · The Boys' Brigade 5th Singapore Company — Dec 2024 – present

- I'm the link between the Boys and the Officers. I'm the Company's only Drill Instructor in 2026 (I passed both the practical and theory assessments). I was second-in-command of Blaze 2026, where I trained three teams for an interschool 21 km adventure race: first aid, knots, orienteering and fitness. I was Assistant Camp Commandant for Adventure Camp 2026 and ran the day-to-day operations, and I'm third-in-command of the Awards Department.

Senior Exco member, SEED (Computer Science Club) · Singapore Polytechnic — 2026

- I organised and taught the "How to COMe Home" Computer Organisation crash course for juniors on 20 May 2026.

Publicity Advisor, Gryphons (Cybersecurity Club) · Singapore Polytechnic — 2026

- I designed the poster for the joint SEED and Gryphons AGM and helped at the Youth Cyber Exploration Camp.

Class Management Committee · Singapore Polytechnic — 2026

- Planned the class outing and passed information between teachers and the class.

AWARDS

- Founder's Award, The Boys' Brigade in Singapore, Jun 2024. It is the highest award in the Senior Programme.
- Best Primer Merit Award, 5th Company, 4 Apr 2026.
- Edusave Certificate of Academic Achievement 2026 (top 25% of my level); MOE EAGLES Award and Edusave Good Progress Award, Nov 2024.
- Black Belt, Dan 1, Singapore Taekwondo Federation.

EDUCATION

Singapore Polytechnic · Diploma in Cybersecurity and Digital Forensics — Apr 2025 – Mar 2028

Distinctions in Digital Forensics and Investigation and Personal Branding and Career Agility. Other modules: Ethical Hacking Essentials, Offensive Security, Linux Administration and Security, Securing Microsoft Windows, Back-End and Front-End Web Development, Programming Methodology, Computer Organisation, AI for Impact. This semester: Malware Reverse Engineering, Applied Cryptography, Data Protection for Cyber Security.

Kuo Chuan Presbyterian Secondary School · GCE O-Level, L1R4: 9 — Jan 2021 – Sep 2024

SKILLS

AI and agents Claude Code, Claude API and Platform, MCP servers, multi-agent setups, guardrails and guard tests.

Build Python and JavaScript (fundamentals); Next.js, React, Supabase. Rust and TypeScript via AI-assisted work I direct and review.

Security Digital forensics, binary exploitation, Linux, Windows hardening, CTFs.

Spoken English, Mandarin.